

I. Objective

To characterize the monodispersity, size, stability, and solubility of TMOS nanogel (NG) to determine whether or not it is a stable precursor for nanogel fabrication.

II. Introduction

Tetramethoxysilane, $\text{Si}(\text{OCH}_3)_4$, is an important silicon alkoxide reagent used in sol-gel method of nanogel fabrication; the goal of this synthesis is to generate a stable, protective nanogel vehicle capable of sufficient nitroxide loading for use as a non-toxic MRI contrast agent. In sol-gel synthesis, TMOS is readily hydrolyzed to a colloidal suspension, and then undergoes polycondensation and gelation. This gel is then processed to a powder, and then analyzed for size, stability, surface charge, and distribution. The alkoxide precursor used in fabrication is relevant – alkoxides react easily with water, encouraging the initial hydrolysis, and the specific alkoxide used affects the overall stability of the nanogel matrix. Ideally, the nanogel generated from the precursor contains colloidal particles of uniform size and remains stable over time.

To evaluate the efficacy of an alkoxide precursor, dynamic light scattering (DLS) is used to characterize the colloid's particle size and monodispersity by analyzing the Brownian motion of suspended particles. The Zetasizer extrapolates this movement (rather, the magnitude of flux in light scattering) to size, and generates a report on particle size, uniformity (monodispersity), and magnitude of monodispersity. It measures stability via the zeta potential, which indicates whether a solution is likely to aggregate, solubilized, or neither based on electric potential of the particle surface (i.e. its charge). Transmission electron microscopy (TEM) can be used alongside DLS to obtain more precise determinations of size and shape of particles.

An ideal nanogel is highly monodisperse ($\text{PDI} < 0.5$), particles averaging 100 nm in size, with a count rate between 100-500 kcps, and a ± 30 mV zeta potential.

III. Procedure

I prepared DLS sample by transferring 60 mg of TMOS NG (sample date 12-19-21) to an Eppendorf tube, and added 2 ml of Picopure water; concurrently I set up my TEM sample. For that, I transferred 40 mg TMOS to an Eppendorf, added 1 ml distilled water, and sonicated both samples for 20 minutes. After, I transferred the supernatant of each into separate beakers, and filtered each solution once with a PVDF 0.45 μm syringe filter. The TEM sample was set aside.

For DLS analysis, I transferred 750 μl TMOS DLS solution to DLS cuvette, and ran the Zetasizer. I did 9 runs for DLS and 6 runs for zeta potential. Results were too variable, so samples were filtered a second time with a PVDF 0.45 μm filter. I prepared two TEM grids – for DLS TEM, I added 5 μl of DLS solution on each side of the grid. For the other, I added 5 μl of the TEM solution (TMOS and distilled water) to each side of the grid and placed both in the fume hood to dry; after drying, they were put aside for later TEM analysis.

IV. Results / Discussion

The DLS sample was initially run after single filtration; the resulting bimodal peaks indicated a need for a second filtration. After double filtration, Zetasizer advised that sample was still too polydisperse. Data indicated an average PDI of 0.34, although trended downward over time. Particle size was within acceptable range at an average of 270 nm, as was the distribution at

96%; the distribution trended downward over time. Zeta potential average was -9.9 mV; standard deviation was 4.3, with %SD at 43%, indicating a relative degree of variability between runs. The considerable variability in count rate also suggested a trend toward instability of the nanogel; Zetasizer advised that sedimentation of the sample was possible. This could be due to multiple factors; instability of the alkoxide in this application, disruption of hydrolysis or condensation reactions due to human error, contamination, or even the high degree of humidity. See VI. Data (last page).

V. Conclusion

This preparation of TMOS NG was too polydisperse and unstable to be an effective precursor for nanogel synthesis.

VI. Data

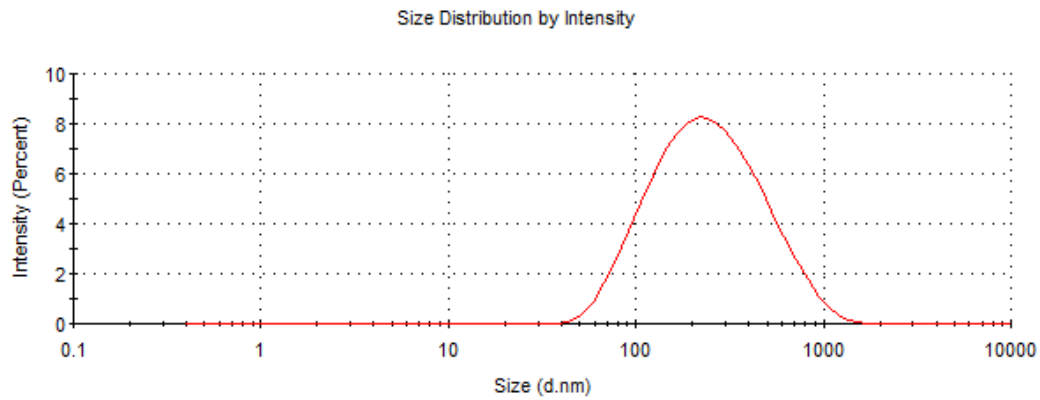


Figure 1 DLS graph of TMOS NG

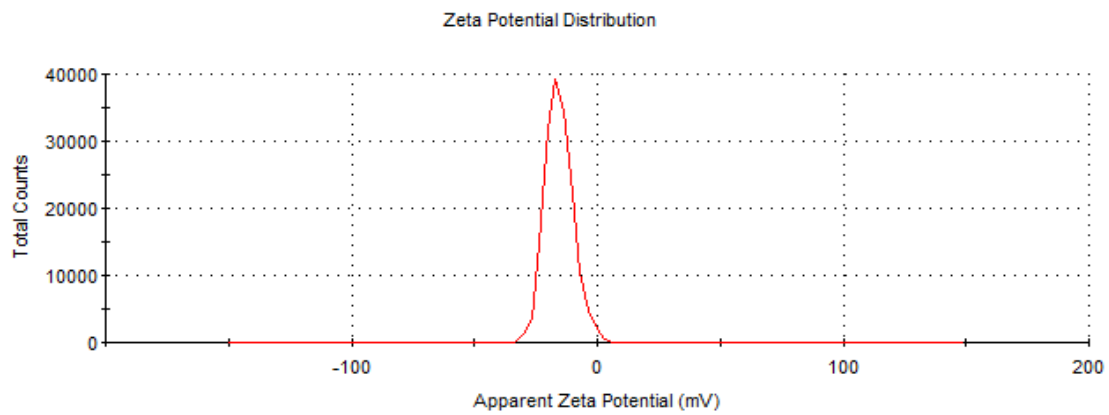


Figure 2 Zeta potential graph of TMOS NG

Table 1 Analysis of TMOS NG					
DLS (nm)	PDI	Distribution (%)	Zeta potential (mV)	Distribution (%)	
380	0.45	100	-2.6		
290	0.37	100	-11		
330	0.38	100	-9.9		
230	0.28	95	-16		
200	0.28	92	-8.8		
200	0.27	88	-12		
270	0.34	96	-9.9		Ave
72	0.074	5.0	4.3		SD
	22%		43%		%SD